

Gesture Recognition Using Edge AI

Problem Statement

Traditional gesture recognition systems rely heavily on cloud-based processing, leading to increased latency, dependency on internet connectivity, and potential privacy concerns. Additionally, implementing real-time gesture recognition on resource-constrained devices like the Arduino Uno presents significant challenges due to limited computational power and memory.

This project addresses these limitations by simulating an **Edge AI-based gesture recognition system** that operates locally, demonstrating how intelligent interaction systems can be built even with constrained hardware using optimized logic and system design principles.

Key Objectives Include

- To design a **real-time gesture recognition system** using sensor inputs
 - To simulate an **Edge AI pipeline** (input → processing → classification → output)
 - To implement **multi-sensor feature extraction** using distance and tilt data
 - To develop a **gesture classification model** using structured logic
 - To incorporate **temporal analysis** for dynamic gesture detection (e.g., swipe)
 - To evaluate system performance using **real-time accuracy tracking**
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Features

- **Multi-Sensor Input System**
Combines ultrasonic distance sensing and potentiometer-based tilt detection
- **Preprocessing & Smoothing**
Reduces sensor noise through data smoothing techniques
- **Gesture Classification Engine**
Detects gestures such as:
 - CLOSE
 - FAR
 - TILT LEFT / RIGHT
 - SWIPE (temporal detection)
- **Priority-Based Decision Logic**
Ensures stable and consistent predictions by resolving conflicts between gestures
- **Simulated AI Confidence Scores**
Mimics probabilistic outputs of real machine learning models
- **State-Based Processing**
Uses a structured flow resembling real-time inference systems
- **Real-Time Performance Metrics**
Tracks:
 - Total predictions
 - Correct predictions
 - Overall accuracy (%)

- **Interactive Output System**

Controls LEDs and servo motors based on detected gestures

Goals

- To demonstrate how **Edge AI concepts can be applied** on low-power systems
 - To bridge the gap between **rule-based systems and machine learning models**
 - To create an **interactive and intuitive human-machine interface**
 - To simulate a **deployable AI system architecture** within a virtual environment
 - To build a foundation for transitioning into **real TinyML implementations**
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Benefits

- **Low Latency** – Processing occurs locally without cloud dependency
 - **Privacy-Friendly** – No external data transmission required
 - **Cost-Effective** – Uses minimal hardware and simulation tools
 - **Educational Value** – Demonstrates core AI/ML system concepts
 - **Scalable Design** – Easily extendable to real hardware and ML models
 - **Interactive Experience** – Engaging and intuitive gesture-based control
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Conclusion

This project successfully demonstrates a **simulated Edge AI-based gesture recognition system** using a structured, pipeline-driven approach. By integrating multi-sensor inputs, preprocessing techniques, temporal analysis, and classification logic, the system replicates the behavior of a real-world AI model within a constrained environment.

Although implemented using rule-based logic, the design closely mirrors actual machine learning workflows, including prediction confidence and performance evaluation. This makes it not only a functional prototype but also a strong conceptual foundation for future enhancements involving **TinyML and real embedded AI deployment**.